David B. Sweasey

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Overview

Passionate and determined computer science graduate with practical experience. Interested in research and implementation of machine learning models, neural networks, and general AI concepts. Pursuing graduate school to expand my knowledge, gain new experience, and hone in on my skills.

Education

North Carolina State University – B.S. Computer Science – Concentration in AI – GPA: 4.0 **Graduated: May 2025**

Minor: Mathematics

Relevant Coursework: Accelerated Deep Learning, Intro to Artificial Intelligence, Data Structures and Algorithms, Data Science, Automated Learning and Data Analysis, Numerical Analysis, Linear Algebra

Honors: Deans List (Fall 2021 – Spring 2025)

Skills

Technologies/Languages: Python, Java, Ruby, C/C++, Javascript, HTML, CSS, React, Node, SQL, Docker, PyTorch, CNNs, Pandas, Numpy, Matplotlib, MATLAB, Excel, Word, Git/GitHub, Windows, Linux on Ubuntu, BeautifulSoup, UiPath, Azure AI Foundry, Power Apps, Power Automate, Power Pages, REST, Websockets.

Soft Skills: Communication, Leadership, Organization, Planning, Collaboration, Problem Solving, Design, Attention to Detail

Projects

Digital Twin– Personal Project **July 2026 – Present**

- An AI digital twin that answers professional questions on my behalf. It handles projects, skills, work, and education.
- Frontend built using JS + React natively within my personal website. Backend implemented with OpenAI SDK, Python, and FastAPI and hosted as a web service through Render.
- Has per day and per session rate limiting as well as content filtering to prevent malicious use.
- Grounded on information from my LinkedIn, resume, and project portfolio. Leverages tools for context lookup, contact.

Intelligent API Scheduler – Personal Project **June 2026 – August 2026**

- Collaborated to develop a full stack application using a “pizza delivery tracker” visual to facilitate the North Carolina Department of IT’s procurement process. Has an email-based authentication system.
- Backend implemented in ruby on Rails, frontend constructed with plain HTML, CSS, and Javascript
- Pulls data from an external database via API calls; incorporates a local MySQL database for temp storage

FLUID – NCDIT **November 2025 – April 2026**

- Agentic solution focused on helping the end user fill out complex forms. It auto-fills basic contact information and helps answer more complex or technical questions.
- Built using Azure semantic kernel for LLM orchestration, Power Pages to display the frontend, Power Automate for integrations between the backend and frontend.
- Collaborated with a small team to prototype, test, and operationalize our solution.

Skin Cancer Neural Network – CSC 422 – Automated Learning and Data Analysis **Oct 2024 – Dec 2024**

- Trained a convolutional neural network on 10,000 images of skin lesions using Python, PyTorch, and Numpy
- Model could differentiate between benign and malignant with 90% accuracy on unseen testing data
- Employed confusion matrices and heat maps to gauge model efficacy

Experience

AI and Automation Analyst – North Carolina Department of IT (NCDIT) **June 2025 – Present**

- Leveraging UiPath and Power Automate to develop, test, and maintain automations for Statewide IT Procurement
- Received Microsoft certification as an Azure AI Engineer Associate. Employing Azure AI Foundry as backend and Copilot Studio + Power Pages as frontend for a procurement form completion assistant.
- Exploring ways to incorporate agentic and/or generative AI solutions in to the Statewide IT Procurement pipeline